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Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China

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Abstract

The spatial extent of humid heat depends on both regional temperature levels and their geographic distribution. We compare spatial contrasts and simultaneous wet-bulb temperature coverage in eastern China over 33 summers in 1991–2025, following development on 2015 and 2022. On high-regional-mean days, the five-scale weighted squared pairwise contrast was 7.28% lower. Contraction increased with spatial bandwidth and was concentrated in the north–south climatological gradient. On a nested 465-site grid, the land fraction exceeding $24°$C rose from 37.39% on middle days to 50.29% on high days; the largest connected fraction rose from 32.10% to 44.67%. Both differences peaked near 23–$24°$C over the full 9.5–$28.5°$C threshold range. In blocked temporal validation, summaries of 121 sites combining mean, variance, latitude gradient and direct exceedance proportion described dense-grid area with a mean absolute error of 0.827 percentage points. Graph and binned-variogram additions gave 0.820 and 0.815 points. The historical extension and station comparison also showed broad-scale contraction. These results connect a weaker geographic gradient to widespread simultaneous humid heat and show that simple spatial summaries retain most coverage information.

Keywords ERA5-Land, graph Laplacian, humid heat, randomisation inference, semivariance, spatial statistics

1. Introduction

Wet-bulb temperature (WBT) is a widely used measure of humid heat stress that combines air temperature and humidity (Sherwood and Huber, 2010; Raymond et al., 2020). Two days can have identical regional means yet sharply different spatial fields: one may contain strong north–south or local contrasts, while the other is comparatively coherent.

Recent work has documented increasingly concentrated humid-heat extremes in standard climate regions (Speizer et al., 2022), sharp local wet-bulb extremes associated with wet soils (Chagnaud et al., 2025), and reduced intra-urban heat-stress variability during some heat waves (Shreevastava et al., 2023). Spatial-extremes models such as Healy et al. (2025) estimate nonstationary temperature tails and the area exceeding critical thresholds. The within-month comparison here examines how pairwise WBT differences change with graph scale. Ordinary cross-site variance combines neighbouring differences, mesoscale structure and the domain-wide gradient in a single value. Empirical variograms represent distance-dependent variation, and kernel smoothing has been used to estimate isotropic semivariograms nonparametrically (García-Soidán et al., 2004). Here, Gaussian weights define a multiscale regime-contrast summary for observed fields across the prespecified distance scales. Smooth weighting provides this summary and permits an exact spatial allocation of the regional contrast.

We ask how the spatial pattern of wet-bulb temperature changes on days with high regional means. In eastern China, humid heat varies along a persistent north–south gradient as well as over shorter distances. A reduction in the regional variance could arise from a weaker geographic gradient, smaller local fluctuations, or both. Distinguishing these changes establishes how regional humid-heat conditions relate to the distribution of spatial contrasts. Topography (Pepin et al., 2015), soil moisture (Seneviratne et al., 2010) and circulation (Horton et al., 2015) provide physical context for these patterns.

The area above a temperature level and the size of its largest connected region describe how extensively humid heat occurs at one time. We measure both quantities across a fixed temperature range and examine their relationship to the spatial contrasts. We also assess how much information about these dense-grid states is retained by summaries from the original 121 sites, comparing graph scales with variance, latitude gradient and binned semivariances.

We use Gaussian-weighted graph semivariances to compare high-mean and middle-mean days at five spatial bandwidths. Graph Dirichlet energy and semivariograms describe weighted pairwise variation (Cressie, 1993; Shuman et al., 2013; Dong et al., 2016). Averaging the relative contrasts equally over months, bandwidths and summers gives a regional summary; retaining the bandwidth profile shows how contraction varies with distance. An exact decomposition assigns each pairwise contribution to its two sites, and Laplacian projections separate the north–south gradient from the remaining spatial variation. Together, these calculations identify the scales and geographic structures associated with high regional WBT.

The analysis uses 121 ERA5-Land sites during 1991–2025. We developed the daily field selection, event definition and bandwidths using 2015 and 2022, then applied them to the other 33 summers. Defining events from the regional mean and fixing the bandwidths before evaluation separates these choices from the spatial contrasts being compared (Simmons et al., 2011; White, 2000). We assess the contrast with record-wise cyclic shifts and examine its sensitivity to event thresholds, field selection and spatial support. The 1950–1990 record extends the temporal comparison, and NOAA observations provide a comparison on a station network. The scale profile and spatial decompositions distinguish attenuation of the climatic gradient from changes in daily anomaly energy. The nested-grid comparison quantifies the accompanying total and connected humid-heat areas and the information retained by simple and distance-resolved spatial summaries.

2. Quantifying spatial contrasts

2.1. Graph dispersion

Let y_{it} denote WBT at location i in the selected field for day t, and let $\bar{y}_t = n^{-1}\sum_i y_{it}$. The benchmark is the site mean squared spatial deviation,

$$V_t = \frac{1}{n}\sum_{i=1}^{n}(y_{it} - \bar{y}_t)^2, \qquad n = 121. \tag{1}$$

To retain spatial arrangement, we use an equirectangular kilometre projection centred at the mean site latitude: longitude is multiplied by $111.32\cos(\bar{\phi})$ and latitude by 110.57. For bandwidth h, set $w_{ij,h} = \exp\{-d_{ij}^2/(2h^2)\}$ for $i \neq j$ and $w_{ii,h} = 0$. Let $\mathbf{W}_h = (w_{ij,h})$, $\mathbf{L}_h = \mathrm{diag}(\mathbf{W}_h\mathbf{1}) - \mathbf{W}_h$, and $S_h = \sum_{i<j} w_{ij,h}$. We define graph dispersion as

$$Q_h(\mathbf{y}) = \frac{\mathbf{y}^\top \mathbf{L}_h \mathbf{y}}{2S_h} = \frac{\sum_{i<j} w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} w_{ij,h}}. \tag{2}$$

Graph dispersion is therefore one half of the edge-weighted mean squared pairwise difference, equivalently an edge-weighted semivariance. Dividing by S_h makes the measure comparable across graphs with different total edge weights. Smaller values indicate greater graph-scale coherence. It has units $^\circ\mathrm{C}^2$; $D_h^{\mathrm{RMS}} = (2Q_h)^{1/2}$ is the weighted root mean squared pairwise WBT difference in $^\circ\mathrm{C}$.

Equation (2) connects three familiar descriptions of spatial variation. It is the graph Dirichlet energy used to measure graph signal roughness (Shuman et al., 2013; Dong et al., 2016). If the field is second-order stationary with semivariogram $\gamma(d_{ij}) = \frac{1}{2}\,\mathrm{E}(Y_i - Y_j)^2$, then

$$\mathrm{E}\{Q_h(\mathbf{Y})\} = \frac{\sum_{i<j} w_{ij,h}\gamma(d_{ij})}{S_h}, \tag{3}$$

so its expected profile is a kernel average of the semivariogram. Under second-order stationarity, this identity supplies a semivariogram interpretation. We calculate Q_h directly from each observed WBT field. On a complete graph with equal off-diagonal weights, $Q_h = nV/(n-1)$. Ordinary spatial variance is therefore the complete-graph endpoint of the same family, up to its degrees-of-freedom factor.

We use five bandwidths selected during method development: $h/d_{\mathrm{med}} \in \{0.125, 0.25, 0.5, 1, 2\}$, where d_{med} is the median pairwise distance. Write these physical bandwidths as h_k, $k = 1, \ldots, H$, with $H = 5$; k is a scale index and h_k is a distance in kilometres. They correspond to 126, 252, 503, 1,006 and 2,013 km. Their weighted mean pair distances are 200, 319, 549, 826 and 992 km, and their effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. Thus 126 km denotes a Gaussian bandwidth with no hard distance cut-off; it is local only relative to the sampled support, where the median nearest-neighbour distance is 150 km. For edge weights w_e, the reported Kish quantity is $N_{\mathrm{eff}} = (\sum_e w_e)^2/\sum_e w_e^2$.

Let $r = (y, m)$ identify a year and calendar month, with $m \in \mathcal{J} = \{6, 7, 8\}$. Define the set of retained days $\mathcal{T}_{ym} = \{t : t \text{ belongs to year } y \text{ and month } m\}$. For $p \in (0, 1)$, let $q_{p,ym}$ be the type-7 sample quantile of the collection $\{\bar{y}_t : t \in \mathcal{T}_{ym}\}$. The three disjoint sets are

$$\begin{aligned}
\mathcal{E}_{ym} &= \{t \in \mathcal{T}_{ym} : \bar{y}_t \geq q_{0.75,ym}\}, \\
\mathcal{M}_{ym} &= \{t \in \mathcal{T}_{ym} : q_{0.25,ym} \leq \bar{y}_t < q_{0.75,ym}\}, \\
\mathcal{L}_{ym} &= \{t \in \mathcal{T}_{ym} : \bar{y}_t < q_{0.25,ym}\}.
\end{aligned} \tag{4}$$

Only $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ enter the primary comparison; $\mathcal{L}_{ym}$ denotes the excluded low group. A record subscript r always abbreviates (y, m): for example, $R_{rh} = R_{ymh}$ and $c_{irh} = c_{iymh}$. The thresholds follow each record's regional-mean distribution. We examine within-month seasonal progression through calendar matching. There were no threshold ties; each record has eight high days and 14 or 15 middle days. Quartiles define a relative warm state against the middle half of the monthly distribution, providing eight high days and at least 14 days for the reference mean. Sensitivity analyses replace both cut points, including an upper-tertile versus middle-tertile comparison, while keeping the fields, graph scales and aggregation fixed (Supplement, Section S12).

The two groups are compared at each scale by the relative difference of their dispersion means. Let $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ be the high- and middle-day sets defined above, and let $\overline{Q}^{E}_{ymh}$ and $\overline{Q}^{M}_{ymh}$ denote the corresponding graph-dispersion means, for example $\overline{Q}^{E}_{ymh} = |\mathcal{E}_{ym}|^{-1}\sum_{t\in\mathcal{E}_{ym}} Q_h(\mathbf{y}_t)$, with the middle mean defined analogously. The scale-specific relative contrast and its summer average are

$$R_{ymh} = \frac{\overline{Q}^{E}_{ymh}}{\overline{Q}^{M}_{ymh}} - 1, \qquad \widehat{\theta}_y = \frac{1}{|\mathcal{J}|H}\sum_{m\in\mathcal{J}}\sum_{k=1}^{H} R_{ymh_k}. \tag{5}$$

For a set $\mathcal{Y}$ of summers, the finite-record target is

$$\widehat{\theta}_{\mathcal{Y}} = \frac{1}{|\mathcal{Y}|}\sum_{y\in\mathcal{Y}} \widehat{\theta}_y. \tag{6}$$

A negative value means that high-regional-mean fields vary less across space. Because the bandwidths are equally spaced on the log scale, the arithmetic mean gives equal weight to every bandwidth in the prespecified five-scale grid. The primary summary averages record-specific ratios. Its asymmetry and sensitivity to small middle-day means motivate two further comparisons. The log-ratio analysis replaces R_{ymh} by $\log(\overline{Q}^{E}_{ymh}/\overline{Q}^{M}_{ymh})$ at the same three stages of averaging and back-transforms the final mean for interpretation. A second analysis uses the bounded symmetric contrast $2(\overline{Q}^{E}_{ymh} - \overline{Q}^{M}_{ymh})/(\overline{Q}^{E}_{ymh} + \overline{Q}^{M}_{ymh})$. The three contrast measures are applied to identical fields and labels in the denominator stress test. We treat the log ratio as the principal robustness analysis and the bounded symmetric form as a secondary check.

2.2. Decomposing the spatial contrast

Graph dispersion accumulates over edges. We obtain an exact and symmetric site allocation by splitting every undirected edge equally between its endpoints. For location i, define

$$q_{ih}(\mathbf{y}) = \frac{1}{4S_h}\sum_{j\neq i} w_{ij,h}(y_i - y_j)^2. \tag{7}$$

The identity is exact because the double sum counts every edge twice:

$$\sum_{i=1}^{n} q_{ih}(\mathbf{y}) = \frac{1}{2S_h}\sum_{i<j} w_{ij,h}(y_i - y_j)^2 = Q_h(\mathbf{y}). \tag{8}$$

The allocation is translation invariant and nonnegative for a single field. It also respects the graph geometry: a site receives contributions only from its incident weighted contrasts, while distant pairs carry little weight in a local graph. Equal edge splitting avoids assigning an arbitrary direction to an undirected spatial relationship.

Let $\overline{q}^{E}_{iymh}$ and $\overline{q}^{M}_{iymh}$ denote the high- and middle-day means within record (y, m), and let $\overline{Q}^{M}_{ymh} = \sum_i \overline{q}^{M}_{iymh}$. The site contribution to the record-level relative contrast is

$$c_{iymh} = \frac{\overline{q}^{E}_{iymh} - \overline{q}^{M}_{iymh}}{\overline{Q}^{M}_{ymh}}, \qquad \sum_i c_{iymh} = R_{ymh}. \tag{9}$$

The regime-change quantity c_{iymh} may be negative. Its denominator is the record-specific middle-day dispersion. Sites in months with different baseline variability therefore enter the final allocation on the same relative scale as the corresponding record-level estimator.

Let $\mathcal{Y}$ be the set of summers. With $H = 5$, the reported node allocation is

$$c_i = \frac{1}{|\mathcal{Y}|}\sum_{y\in\mathcal{Y}} \frac{1}{|\mathcal{J}|}\sum_{m\in\mathcal{J}} \frac{1}{H}\sum_{k=1}^{H} c_{iymh_k}, \qquad \sum_{i=1}^{n} c_i = \widehat{\theta}_{\mathcal{Y}}. \tag{10}$$

The order and weights in Equation (10) match the estimator: equal months within a summer, equal graph scales, and equal summers. Changing the spatial support changes the incident edges and hence the allocation. Accordingly, every main-text map is calculated from the 121-node graph. For a scale-specific map, retaining one h in Equation (10) yields the corresponding scale-specific contrast through the observed-node sum.

For display, we fit a low-rank thin-plate REML surface to each set of 121 observed-node values and evaluate it on a fine land-masked raster. Graph construction, estimation, uncertainty calculations and node-sum identities all use the observed nodes, which are overlaid on each map. A negative c_i allocates part of the regional contraction to edges incident to site i; the allocation measures that site's share of the pairwise contrast. The maps also show regime means of $y_{it} - \bar{y}_t$ and their difference. These spatially centred fields separate a change in spatial pattern from a shift in the daily regional mean.

To quantify the dominant north–south structure, let $\boldsymbol{\ell}$ be centred latitude and $\mathbf{z}_t = \mathbf{y}_t - \bar{y}_t\mathbf{1}$. Define

$$\widehat{\beta}_{th} = \frac{\boldsymbol{\ell}^\top \mathbf{L}_h \mathbf{z}_t}{\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}, \qquad \mathbf{e}_{th} = \mathbf{z}_t - \widehat{\beta}_{th}\boldsymbol{\ell}.$$

Orthogonality in the $\mathbf{L}_h$ inner product gives

$$Q_h(\mathbf{y}_t) = \frac{\widehat{\beta}^2_{th}\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}{2S_h} + \frac{\mathbf{e}^\top_{th}\mathbf{L}_h\mathbf{e}_{th}}{2S_h}. \tag{11}$$

High-minus-middle means of the two terms, divided by middle-day total dispersion, add exactly to R_{ymh}. More generally, for a centred spatial basis matrix $\mathbf{B}$, the Laplacian projection has coefficients $\widehat{\boldsymbol{\beta}}_{th} = (\mathbf{B}^\top\mathbf{L}_h\mathbf{B})^{-}\mathbf{B}^\top\mathbf{L}_h\mathbf{z}_t$, where $(\cdot)^{-}$ is a generalised inverse. Centred latitude is the reference structural summary. Further projections add longitude and then elevation, obtained by dividing the official invariant ERA5-Land geopotential field by standard gravity. We compare the total structured and residual energies of these nested column spaces.

A second decomposition separates the persistent monthly field from daily departures. Write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$, where $\boldsymbol{\mu}_m$ is the site-specific 35-summer mean field for month m. Then

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \tag{12}$$

The first term is constant within a month and cancels from the high-minus-middle numerator. The other two terms quantify change in anomaly energy and change in climatology–anomaly alignment. Dividing both differences by the middle-day total dispersion yields two components that add exactly to R_{ymh}. The signed cross term measures alignment between the monthly climatology and daily anomalies; a negative value indicates opposing spatial patterns.

3. Inference for the observed summers

3.1. Randomisation by cyclic shifts

Days within a month are serially dependent, so each month-year record is randomised as a unit. Its complete daily outcome profile is rotated by a randomly chosen offset while the mean-WBT series and labels remain in place. This joint rotation preserves dependence between bandwidths, and the statistic is recomputed after every shift. The one-sided Monte Carlo p-value is

$$p = \frac{1 + \sum_{b=1}^{B} I(T_b^{\text{shift}} \le T_{\text{obs}})}{B+1}. \tag{13}$$

Validity requires cyclic invariance, a stronger property than generic stationarity. For record r, let $X_r = (\bar{y}_{rt})_{t=1}^{n_r}$ denote the mean series and let D_r collect the H graph-dispersion values for each day, with $X = (X_1, \ldots, X_R)$ and $D = (D_1, \ldots, D_R)$. The operator C_{n_r} rotates all H columns of D_r together, and $G = \prod_r C_{n_r}$ is the group of all joint rotations. The null hypothesis is group invariance:

$$(D \mid X) \overset{d}{=} (gD \mid X) \quad \text{for every } g \in G. \tag{14}$$

Conditional on the observed regional-mean series, independently changing the circular phase of the complete multiscale dispersion profile in any month-year record must leave the joint distribution of all profiles unchanged. Under this null, the group-rank argument gives a finite-sample-valid lower-tail test; the proof and the plus-one Monte Carlo result are in Supplementary Section S1 (Phipson and Smyth, 2010).

Truncated time-shift tests replace wrap-around invariance with strict stationarity (Harris, 2020; Yuan and Shou, 2024), but their conservative finite-sample form requires at least 19 shifts on each side for a 0.05 cutoff. Records of 30 or 31 days leave no usable aligned window under that requirement, so we use cyclic shifts under Equation (14).

The product condition is stronger than marginal cyclic invariance of each record because it permits the records to be rotated independently. It specifies the conditional distribution of the full profiles, beyond their mean contrast. Common circulation or soil-moisture states across adjacent months, seasonal progression and large-scale climate drivers can violate independent phase invariance. A targeted simulation in Supplementary Section S11.4 studies one shared cross-record process and its effect on test size. Scale-wise weak family-wise error control and the development-record summaries are also detailed in Supplementary Section S1.

3.2. Development and evaluation samples

We used the summers of 2015 and 2022 to inspect WBT fields and graph profiles and select the daily fields, event labels, spatial support, bandwidths and aggregation. The remaining 33 summers form the evaluation record $\mathcal{Y}_H$ in Equations (5)– (6). The available records identify the two development summers but leave their original acquisition rationale undocumented.

For the global cyclic-shift analysis, set $T_{\text{obs}} = \widehat{\theta}_{\mathcal{Y}_H}$. Each Monte Carlo transformation rotates the five-column dispersion profile within each of the 99 evaluation month-year records, then recomputes every record ratio and the equal-month, equal-scale and equal-summer mean, with $B = 99{,}999$.

The exploratory global cyclic-shift analysis followed estimation of the 33-summer contrast. The statistic is the evaluation-record target in Equation (6). The test uses the conditional product-invariance null in Equation (14).

Contrast magnitude is reported through the record mean, summer-specific contrasts, negative-summer count and leave-one-summer-out range.

Version 2 of the analysis protocol is dated 2 August 2026. It specifies the Student, lag-2 Newey–West and sign-test components and their joint decision rule. Supplementary Table S1 gives the dates, statistical assumptions and results of the primary and subsequent analyses.

3.3. Inference under temporal dependence

For comparison, a process-level reference treats the summer sequence as a stationary weakly dependent process. Under standard strong-mixing, moment and long-run variance conditions, its mean admits a central-limit approximation; a growing-bandwidth HAC estimator needs a stronger uniform autocovariance condition (Ibragimov and Linnik, 1971; Newey and West, 1987; Andrews, 1991). The full statement, ratio negative-moment conditions and proof are in the Supplement. At 33 summers, simulation shows that Student and HAC intervals can have substantial undercoverage under serial dependence. Both intervals are reported alongside the finite-record randomisation result.

4. Simulation study

The simulations examine day-level cyclic randomisation and the behaviour of process-model intervals at the 33-summer record length. The number of summers determines the repeated-summer sample size, whereas day-level simulations address within-record randomisation. Complete data-generating specifications and result tables are in Supplementary Section S11.

4.1. Simulation design

The day-level experiment uses the observed coordinates and six 30–31-day records. Its circular Gaussian null satisfies the invariance condition in Equation (14); two finite-autoregression variants deliberately violate wrap-around invariance. Each Gaussian anomaly is projected through $\mathbf{I}-\mathbf{1}\mathbf{1}^\top/n$, so its sample mean across the 121 sites is exactly zero and the field mean equals the series used to form labels. The centred anomaly is independent of that mean. This calibration design isolates the randomisation properties; the fixed-hour and anomaly-field analyses in Section 5 examine field selection and label–field dependence in the observed record. Amplitude reduction, correlation-range expansion and gradient suppression produce true profile contractions of 5.9%–27.4%, calculated from the known quadratic forms. The Monte Carlo standard error is 0.7 percentage points at 5% rejection and 1.6 points at 50% power, so differences of one or two points are difficult to distinguish.

The comparison includes graph and binned-variogram profiles, spatial variance, two local semivariances, Moran's I and Geary's C. These comparisons cover the specified amplitude, range and gradient alternatives. The repeated-summer experiment compares Student, fixed-lag and growing-lag calculations, the sign test and the protocol's three-component rule.

4.2. Calibration and power

Figure 1 summarises the power ranges over nine spatial alternatives; complete values are in the Supplement.

Across eight cyclic-invariant designs, graph-profile rejection ranged from 4.5% to 6.8%, while the six competitors spanned 4.0%–6.4%. The graph test had rejection rates of 4.5% at temporal correlation 0.75 and 6.1% under the t_3 field. The two non-circular boundary cases gave 4.6% and 5.1%.

Gaussian-alternative absolute bias was below 0.48 percentage points and root mean squared error was 2.6–3.8 points. Under the original t_3 null, graph randomisation size was 6.1%. A paired denominator stress test then used 2,000 heavy-tailed data sets under each of the null and -7% alternatives and 999 common product shifts per data set. Null rejection was 4.70%, 4.90% and 4.75% for the raw ratio, log ratio and bounded symmetric contrast. On their native scales, the corresponding null absolute biases were 0.372, 0.070 and 0.066, and root mean squared errors were 1.472, 0.328 and 0.277. Under the -7% alternative, root mean squared errors were 1.369, 0.328 and 0.275, while rejection

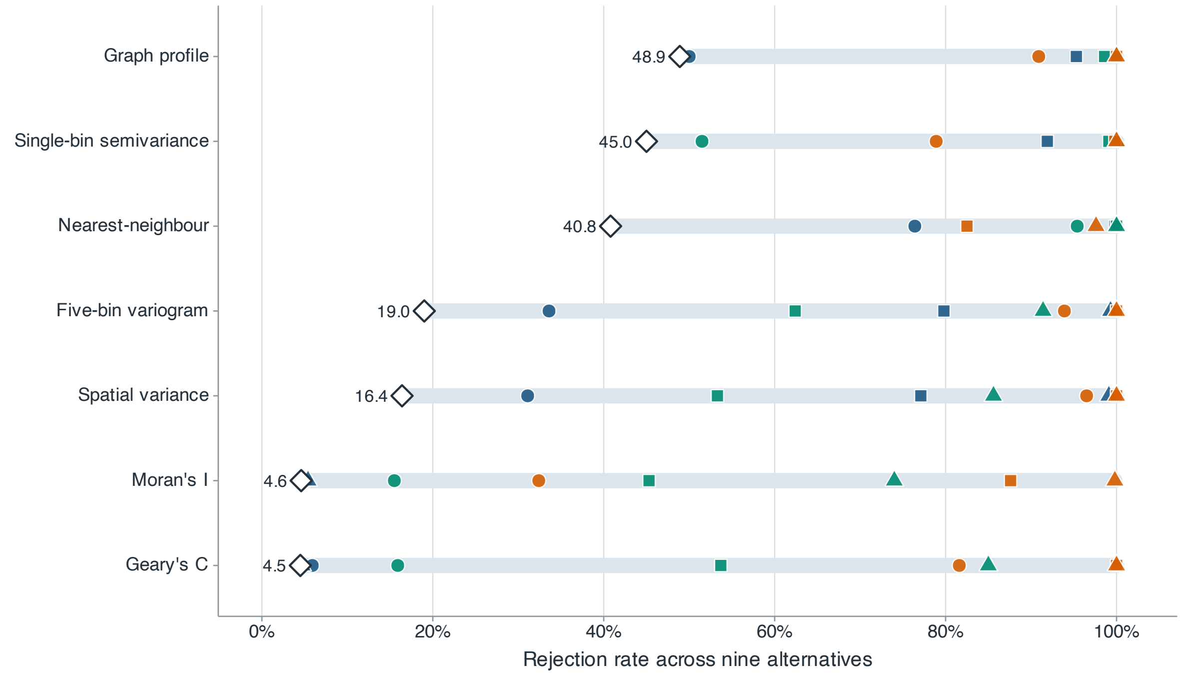

Figure 1 Power across nine simulated spatial alternatives. Blue, green and orange marks denote amplitude reduction, correlation-range expansion and gradient suppression; circles, squares and triangles denote weak, medium and strong changes. For each method, the pale segment spans power across the nine alternatives and the diamond marks the minimum.

Alt text: A range-and-dot plot compares seven methods. Each method has nine coloured and shaped points, a pale segment spanning their range and a diamond at the minimum; the ranges overlap substantially.

was 7.95%, 8.50% and 7.90%. The bounded contrast had sharply lower estimator dispersion, with similar power in this design. Randomisation calibration can coexist with unstable effect estimation; resistant summaries reduce the instability while leaving residual estimator variation.

Performance varies across the nine simulated mechanisms. Nearest-neighbour semivariance is strongest for range changes, but its power is 40.8% under weak gradient suppression, compared with 90.9% for the graph profile and 96.5% for variance. The five-bin variogram is similar to the graph profile for gradient suppression (93.9% versus 90.9% at the weakest setting), but is less sensitive to the weak amplitude and range changes (33.6% and 19.0% versus 50.0% and 48.9%). Moran's I and Geary's C have near-nominal power for pure amplitude changes because standardisation removes amplitude. The profile's minimum power across the nine alternatives was 48.9%, higher than the minimum of each comparator within this experiment only.

The scale-specific profiles retain information lost by a scalar test. Strong amplitude reduction changes all three summaries by about 19%; range expansion acts most strongly on the local graph (28.2%), while gradient suppression acts on variance and the broad graph (43.9% and 42.5%) and changes the local graph by 9.3%.

Eight additional scenarios allow the label-generating mean and spatial field to share structure; full specifications are in the Supplement. In the shared-latent null, the sign of the spatial gradient has mean correlation 0.61 with regional WBT, but its squared energy is constant, so the true dispersion contrast is zero. The other nulls add a within-month trend, anisotropic spatial covariance or selection of a daily peak from 24 simulated hourly regional means, with hour-invariant anomaly covariance. Corresponding alternatives attenuate the gradient, impose a seasonally declining gradient or reduce high-day anomaly amplitude. Graph null rejection is 4.0%–7.0%, compared with 4.3%–7.3% for the five-bin variogram. Graph rejection under these four nulls is close to its cyclic-invariance calibration. The graph profile is at least as powerful in three of four alternatives, whereas the variogram is slightly stronger for the seasonal-gradient alternative. Smooth graph weighting and exact allocation therefore show no consistent power penalty across these scenarios.

4.3. Interval coverage under serial dependence

At $R = 33$, increasing Gaussian year correlation from 0 to 0.6 reduces Student coverage from 95.2% to 66.1%, fixed-lag coverage from 93.3% to 78.7%, and growing-HAC coverage from 91.2% to 79.0%. Corresponding Gaussian null rejection rises to 21.2%, 14.4% and 14.1%. The error arises mainly from long-run variance estimation. Coverage from all three process-model intervals deteriorates at this record length under the simulated dependence. Full results for 108 cells, including skewed and heavy-tailed innovations, appear in the Supplement.

An additional 10,000-replication experiment uses the exact 1991–2025 calendar and removes 2015 and 2022. The largest absolute mean bias over correlations 0, 0.3 and 0.6 and effects 0 and -7% is 0.03 percentage points. Excluding the two design summers therefore has negligible first-order centring bias, but short-record dependence remains: at correlation 0.6, null coverage is 68.3% for the Student interval that treats summers as independent and 80.2% for calendar lag 2.

5. Humid heat in eastern China

We apply graph dispersion to synchronous wet-bulb temperature fields in eastern China during 1991–2025 and extend the analysis to 1950–1990.

5.1. Constructing the daily fields

ERA5-Land supplies hourly land-surface fields on a 0.1° latitude–longitude grid (Muñoz-Sabater et al., 2021). We use 2-m air temperature, 2-m dew-point temperature and surface pressure for June–August 1991–2025 over 105–125°E and 20–42°N. The rectangle provides a regular support spanning the subtropical-to-temperate north–south gradient of eastern China while excluding most of the Tibetan Plateau west of 105°E. Its northern and southern limits bound that gradient. The analysis panel begins in 1991.

The historical extension covers all 41 June–August seasons from 1950, the first year of the ERA5-Land record, through 1990. It uses the same sites, field construction and five-scale estimator. We report separate summaries for 1950–1978 and 1979–1990 because the earlier segment uses the preliminary ERA5 back extension (ECMWF production documentation). The extension plan and its 99,999-draw product cyclic-shift calculation were recorded before the historical values were retrieved.

Every 16th longitude and 18th latitude cell from a fixed north-west origin gives a 13×13 candidate lattice; removing ocean cells leaves 121 sites (Figure 2). The 35 summers contain 9,350,880 quality-controlled hour–site records and 92 complete UTC fields per summer. Because of NetCDF packing, 2,092 dew-point values exceeded air temperature slightly (0.022% of records); these pairs were clipped at equality before WBT was calculated. The main target is a site average. For area weighting, $a_i = \cos(\text{latitude}_i)$,

$$\bar{y}_t^{(a)} = \frac{\sum_i a_i y_{it}}{\sum_i a_i}, \qquad Q_{h,a}(\mathbf{y}) = \frac{\sum_{i<j} a_i a_j w_{ij,h} (y_i - y_j)^2}{2 \sum_{i<j} a_i a_j w_{ij,h}}.$$

The first area-weighted calculation replaces Q_h by $Q_{h,a}$ using the original labels; the second redefines the labels from $\bar{y}_t^{(a)}$ at the regional peak hours. The two calculations separate spatial weighting from event relabelling.

WBT is calculated from temperature T and dew point T_d in kelvin and surface pressure p in pascals. Vapour pressure is $e(T_d) = 611.2 \exp\{17.67(T_d - 273.15)/(T_d - 29.65)\}$ Pa, and the Bolton mixing-ratio, lifting-condensation-temperature and equivalent-potential- temperature relations are evaluated in these units (Bolton, 1980). Saturated T_w is obtained by solving

$$\theta_e(T_w, T_w, p) = \theta_e(T, T_d, p) \tag{15}$$

at the original pressure with 40 bisection iterations on $[T_d, T]$; even a 100-K initial interval then falls below 10^{-9} K. Values with $T_d > T$ are clipped at equality before evaluation. The Stull approximation provides a comparison (Stull, 2011); the main calculation follows the pseudoadiabatic definition discussed by Davies-Jones (2008).

For each UTC day, the 121 simultaneous values at the peak regional mean form the daily field. Within each month-year, the upper regional-mean quartile is high, the middle half is middle and the lower quartile is excluded. Labels depend on regional mean WBT, and the selected hour maximises that mean; graph dispersion enters after

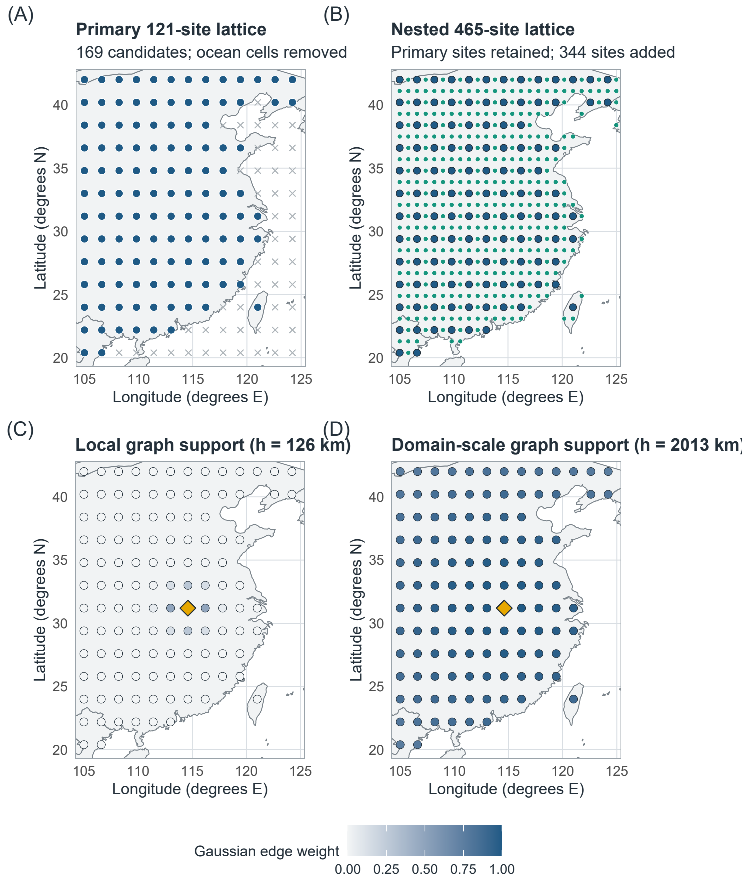

Figure 2 Spatial sampling and graph support. (A) Analysis lattice with 121 land sites from 169 candidates. (B) Nested lattice adding 344 land sites while retaining every analysis site. (C,D) Gaussian edge weights from one reference site under the 126- and 2,013-km graphs, showing how bandwidth changes the geographic support at fixed coordinates. Basemap: Natural Earth 1:50m, accessed through the R `maps` package.

Alt text: Four eastern-China maps compare the analysis and nested site lattices, then show weights concentrated near one reference site under the 126-km graph and spread across the domain under the 2,013-km graph.

selection. UTC+8 days, alternative thresholds, sitewise maxima, all 24 fixed UTC hours and daily-mean fields provide comparisons. At each fixed hour, both the original peak-based labels and labels recomputed from that hour's regional mean are analysed; daily-mean labels are likewise recomputed from daily-mean WBT. A dated event manifest records the type-7 thresholds, membership and absence of ties.

Three checks address within-month seasonal progression: comparison of the day-of-month distributions, removal of a separate linear day-of-month trend at every site within each year–month, and subtraction of each site's calendar-day climatology, estimated from all summers except the one being analysed. The last two calculations use the original event labels. A date-matched calculation compares each high day with all middle days in the same month-year lying within ± 3 or ± 5 calendar days. It first averages the available high-day-specific ratios and then uses the same equal-month, equal-scale and equal-summer hierarchy as the estimator; unmatched-day and empty-record counts are retained.

A continuous diagnostic regresses $\log Q_h$ on centred regional mean WBT within each month-year and bandwidth. Slopes are then averaged equally over months, bandwidths and evaluation summers. This slope supplements the quartile target with a graded summary.

Alternative graph kernels are $\exp(-d_{ij}/\rho)$ and $[1-d_{ij}/\rho]_+^2$. For each Gaussian scale, ρ is found by bisection so that the alternative kernel has the same edge-weighted mean pair distance. Across the five scales, their effective edge counts range from 334 to 7,112 and from 382 to 7,113, compared with 366 to 7,109 for the Gaussian graphs; the full matched-distance and effective-edge metadata accompany the analysis.

A nested $0.8^\circ \times 0.9^\circ$ lattice adds 344 land sites while retaining all 121 locations. Numerical convergence is assessed at the five physical bandwidths by comparing 121 sites with a 239-site longitude refinement, a 236-site latitude refinement and all 465 sites under the evaluation events; peak times and labels are also reselected on the full grid. Thin-plate REML surfaces (Wood, 2003) are used only for display in Figures 3 and 5; every estimate uses observed nodes.

Additional spatial-support checks use WGS84 geodesic distances on the 121 sites and the 465-site lattice. On the latter, each boundary is moved inward by one or two lattice steps and event labels are recomputed from the retained domain. A non-rectangular support intersects the downloaded grid with the Natural Earth low-resolution mainland China feature, excluding the separately stored Taiwan feature. The available fields cover the original rectangle, so the boundary comparisons use inward movements. These checks retain the five physical bandwidths and report complete scale curves.

We compare ERA5-Land WBT with observations from the National Oceanic and Atmospheric Administration Integrated Surface Database (NOAA ISD) (Smith et al., 2011). The station comparison uses ten summers (1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025) selected before station retrieval. Archive, operating-period and geographic eligibility rules followed by a deterministic maximin design selected 30 dispersed stations. A station-year was retained when it had at least 400 quality-controlled, complete exact-hour observations of temperature, dew point and sea-level pressure; station elevation supplied surface pressure under a standard atmosphere, $p_s = p_{\mathrm{SL}}\{1 - 2.25577 \times 10^{-5} z\}^{5.2559}$ for elevation z in metres. At two offshore stations, all variables were missing from the nearest ERA5-Land grid cells, leaving 28 usable stations and 175,172 exact-hour WBT pairs.

The station-field calculation uses peak times and high–middle labels from the 121-site ERA5-Land fields. At each retained time, NOAA and ERA5-Land graph contrasts use the identical station subset and require at least 10 stations. All 30 year-month records retained at least one field in each regime. The 503-, 1,006- and 2,013-km graphs were designated as station-supported before evaluation; the two narrower graphs diagnose the limit of the station geometry. The comparison assesses measurement agreement and the direction of spatial contrast on this sparse network. The two sources are compared at common event times on matched station subsets.

5.2. Measuring simultaneous coverage

We measure the spatial extent of humid heat on the 465-site lattice at the original 121-site peak time. Each retained site represents a $0.8^\circ \times 0.9^\circ$ cell clipped to the study rectangle and the Natural Earth land polygon. WGS84 cell areas sum to 3.221 million km^2, covering 95.7% of the mapped land in that rectangle. Let a_i be the represented land area and $a_+ = \sum_i a_i$. At threshold u, define

$$A_t(u) = \frac{\sum_i a_i I(y_{it} > u)}{a_+}, \qquad C_t(u) = \frac{\max_{B \in \mathcal{B}_t(u)} \sum_{i \in B} a_i}{a_+}, \tag{16}$$

where $\mathcal{B}_t(u)$ contains connected components of exceedance cells under four-neighbour grid adjacency, with $C_t(u) = 0$ when the exceedance set is empty. Multiplication by a_+ gives the corresponding areas in km^2. These quantities describe simultaneous coverage at the sampling resolution. Spatial extent and contiguity are also central to temperature-event analyses by Healy et al. (2025) and Skinner et al. (2025).

The temperature range uses the 1st and 99.5th percentiles of the two development summers, rounded outward to 0.25°C steps. We retain all 77 thresholds from 9.5 to 28.5°C. For each threshold we compare high and middle days within month-year, then average months and evaluation summers equally. The differences are expressed in area percentage points and km^2. Pointwise descriptive intervals resample five-year calendar blocks of the annual contrasts, retaining the positions of the two excluded development years. Eight-neighbour connectivity and cosine-latitude weights provide separate sensitivity comparisons.

We then compare how summaries of the 121-site field describe coverage on the 465-site grid. A mean-and-season regression is followed by a strong simple baseline that also includes spatial variance, the signed latitude slope and the area-weighted 121-site exceedance proportion. Two further models add five binned semivariances or the five graph values, each divided by spatial variance and log-transformed to describe relative scale structure. All four use the same spline regression with ridge regularisation and fitted proportions clipped to $[0, 1]$. Linear spatial interpolation, followed by coverage calculation, provides a full-field comparison.

Seven held-out five-year calendar blocks and one-year training buffers separate evaluation from fitting. Nested blocked validation selects the ridge penalty using training years, following the dependence-aware approach to model assessment discussed by Roberts et al. (2017). Errors are averaged across thresholds, then equally across months and summers, for all days and for high days separately. An additional exceedance comparison uses only the 344 new sites. The connected-area calculation uses all 465 sites. This comparison measures spatial information within ERA5-Land across omitted years. Supplementary Section S13 gives the complete definitions and fitting procedure.

5.3. Stronger contraction at broader scales

Across the 33 evaluation summers, the five-scale contrast on the weighted squared-difference scale was -7.28%. The three-component consistency rule was satisfied. The one-sided Student and lag-2 Newey–West reference values were 2.56×10^{-6} and 3.54×10^{-6}; the independent-summer exact binomial sign reference value was 3.31×10^{-5}. All three were below the prespecified 0.025 threshold. The corresponding two-sided Student and lag-2 HAC reference intervals were $[-10.00, -4.57]\%$ and $[-10.05, -4.51]\%$. The dependence simulations in Section 4 quantify the coverage of these intervals under interannual correlation.

In the exploratory cyclic-shift test, no sampled shift was as negative as the observed statistic. With $B = 99{,}999$ draws, an exceedance count of zero and seed 20260809, the plus-one Monte Carlo value was $p_{\mathrm{MC}} = 1.0 \times 10^{-5}$, the minimum attainable value for this run. The complete five-scale daily profile was shifted jointly within each of the 99 month-year records. The test evaluates the conditional product-invariance null in Equation (14).

Twenty-eight summer-specific contrasts were negative, and leave-one-summer-out estimates ranged from -7.69% to -6.64%. The Student interval above is a descriptive t-reference interval based on between-summer variation. Its calculation assumes independent summers; serial dependence affects its coverage, as shown in Section 4. The fraction negative is 0.85, with a binomial Wilson reference interval of $[0.69, 0.93]$ under independent summer signs and a one-sided independent-summer sign reference value of 3.3×10^{-5}.

The fitted trend was -0.22 percentage points per year (95% descriptive OLS interval based on between-summer variation, $[-0.48, 0.05]$, two-sided $p = 0.10$). The 2008–2025 mean was 2.95 points more negative than the 1991–2007 mean, with a descriptive Welch interval based on between-summer variation of $[-8.43, 2.52]$ points. Both intervals include zero, leaving the temporal trend uncertain.

Contrasts increase in magnitude from -2.86% as the Gaussian bandwidth is $h = 126$ km to -13.27% at $h = 2{,}013$ km (Table 2; Figure 3). Negative-summer counts rise from 22 to 30. A 31-point log-spaced bandwidth profile decreases at each successive evaluated bandwidth across the same h = 126–2,013-km range. Complete-graph variance falls 14.45% and is negative in all 33 summers. The variance summary captures the overall contraction, while the bandwidth profile shows its concentration at broader scales. Distance-binned variograms describe variation over distance intervals; Gaussian weighting gives smooth scale profiles whose site contributions sum to each regional contrast.

Evidence from 1950 to 1990

The 1950–1990 extension gave a five-scale contrast of -11.04% (descriptive t-reference interval based on between-summer variation, $[-12.84, -9.24]\%$). Forty of 41 summer contrasts were negative, and the leave-one-summer-out range was -11.33% to -10.62%. Its 99,999-draw product-shift calculation produced no statistic as negative as the observed value (plus-one $p_{\mathrm{MC}} = 1.0 \times 10^{-5}$, the minimum attainable value for that run, under its conditional product-invariance null). The scale profile was $-5.96, -7.51, -10.08, -14.50$ and -17.17% from 126 to 2,013 km, with 38, 39, 40, 41 and 41 negative summers. The negative sign and strengthening contraction with bandwidth also appear in the 41 earlier summers.

The production-history split gave -10.50% in 1950–1978 (28 of 29 summers negative) and -12.34% in 1979–1990 (12 of 12 negative). The segments are reported separately because the pre-1979 record uses different forcing. The historical climatology–anomaly decomposition re-estimates the site-specific monthly climatological field from the 41 summers in 1950–1990. At 2,013 km, the historical anomaly-energy component was -0.28 points (95% descriptive (t)-reference interval based on between-summer variation, $[-1.01, 0.45]$), whereas the climatology–anomaly cross

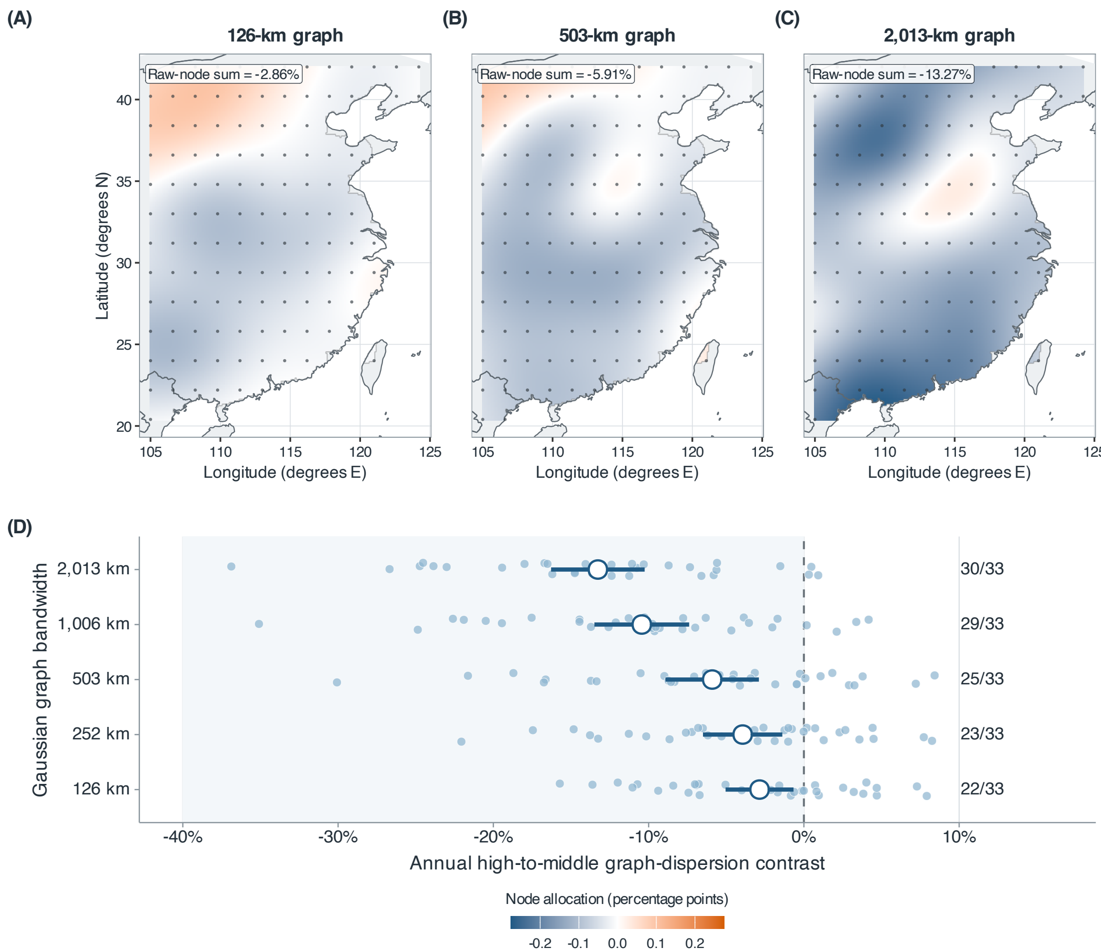

Figure 3 Scale-specific node allocations and summer contrasts for the 33 evaluation summers. (A–C) Allocations for the 126-, 503- and 2,013-km graphs share a symmetric colour scale; their observed-node sums are -2.86%, -5.91% and -13.27%. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes. (D) Small circles show summer-specific contrasts at all five bandwidths; large circles and bars show means and 95% descriptive (t)-reference intervals based on between-summer variation. Labels give the numbers of negative summers.

Alt text: Three smoothed eastern-China maps, each overlaid with the 121 analysis sites, show increasingly coherent negative node allocations at 126, 503 and 2,013 km. A lower panel shows the 33 annual contrasts at all five bandwidths, with mean contrasts and negative-summer counts increasing in magnitude with bandwidth.

component was -16.89 points ($[-19.31, -14.47]$) and negative in all 41 summers. Both the latitude and latitude–longitude structured components were negative in every summer at every bandwidth. In the earlier record, most broad-scale contraction was likewise in the alignment term. The 1950–1978 estimates inherit the uncertainty of the preliminary forcing data.

5.4. Wider coverage on days with high regional means

High-regional-mean days have greater simultaneous exceedance area and larger connected exceedance regions throughout the evaluated temperature range (Figure 4). At 24°C, the mean exceedance proportion rises from 37.39% on middle days to 50.29% on high days. The 12.91-percentage-point difference corresponds to approximately 416,000 km^2 (pointwise 95% block-resampling interval: 377,000–459,000 km^2). The largest connected region rises from

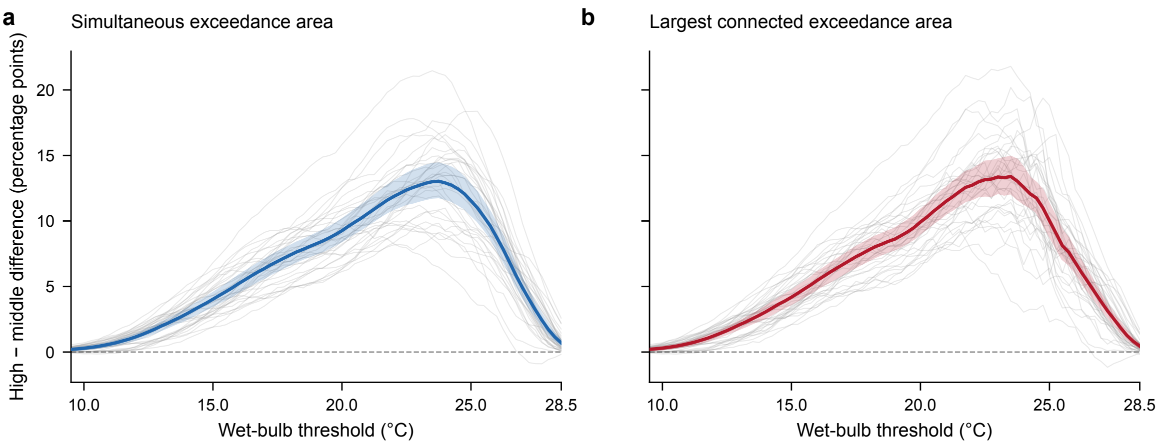

Figure 4 High-minus-middle differences in simultaneous humid-heat coverage at the original daily peak times. (a) Total exceedance area. (b) Largest four-neighbour connected exceedance area. Thick curves average months and 33 evaluation summers equally; thin grey curves show every summer. Shading gives pointwise 95% percentile intervals from 1,999 resamples of five-year calendar blocks. All 77 development-defined temperature thresholds are shown. Areas refer to the 3.221-million-km^2 represented land support.

Alt text: Both area-difference curves rise from small positive values at low thresholds to approximately thirteen percentage points near 24 degrees Celsius, then decline toward the upper end of the threshold range. Grey annual curves show variation between summers, including some negative values at high thresholds.

32.10% to 44.67% of represented land, a difference of 12.57 points or approximately 405,000 km^2 (365,000–452,000 km^2). Both differences are positive in all 33 summers at this threshold.

The mean area difference reaches 13.03 points at 23.75°C; the connected-area difference reaches 13.40 points at 23.5°C. At 28°C the respective differences are 1.72 and 1.27 points, with one summer giving a negative contrast. Thus the broad spatial contraction coincides with larger simultaneous and connected humid-heat areas, with the largest absolute differences around 23–24°C. These group comparisons combine changes in regional mean and spatial pattern. The information comparison below conditions on the available mean and simple spatial summaries.

The strong simple baseline retains most of the coverage information (Table 1). Its mean absolute error is 0.827 percentage points for exceedance area and 1.389 points for the largest connected area, compared with 2.794 and 3.070 points for mean and season alone. Adding the graph profile reduces these errors by 0.0069 and 0.0100 points, respectively. The corresponding binned-variogram model gives errors of 0.815 and 1.366 points, slightly below the graph model's 0.820 and 1.379 points. On high days, graph information reduces the strong baseline errors by 0.0041 and 0.0036 points. Restricting the area reference to the 344 new sites gives the same overall ordering. Graph and variogram gains vary over the temperature range (Supplementary Figure S3). The principal contribution of the graph analysis here is its scale profile and spatial decomposition; the simpler summaries already describe coverage accurately.

5.5. Locating the contraction

The decomposition in Equation (11) localises most of the contrast to the north–south gradient. At 126 km it accounts for -2.58 points (95% (t)-reference interval under an independent-summer model, $[-2.92, -2.24]$) of the -2.86-point contrast. From 252 to 2,013 km its mean contribution is more negative than the total ($-4.60, -8.38, -12.73, -14.96$ points), while residual contrasts contribute $+0.66, +2.48, +2.29, +1.69$ points. The gradient component is negative in all 33 evaluation summers at every bandwidth. Adding longitude to the basis makes the 2,013-km structured component -16.23 points, compared with -14.96 points for latitude alone; the corresponding residuals are positive. The latitude–longitude–elevation basis gives a 2,013-km structured component of -14.34 points (95% descriptive (t)-reference interval based on between-summer variation, $[-16.75, -11.92]$), negative in all 33 summers, and a residual of $+1.07$ points. At 126 km, however, its structured component is -1.16 points (95% descriptive (t)-reference interval based on between-summer variation, $[-2.66, 0.33]$) and is negative in

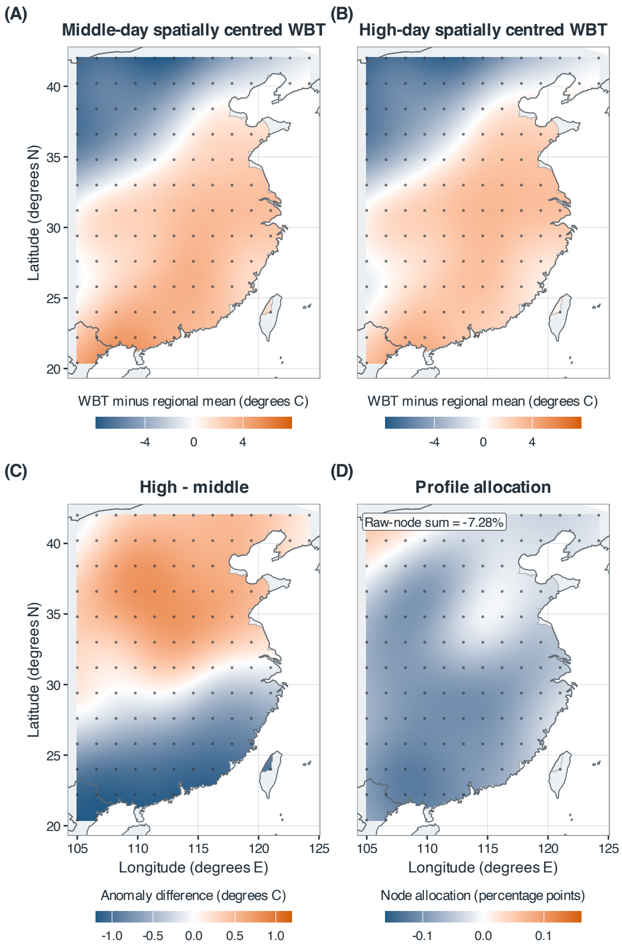

Figure 5 Spatial structure of the 121-site evaluation contrast. (A,B) Middle- and high-day spatially centred fields; (C) their difference; (D) exact observed-node allocations averaged over the five graph scales. The 121 allocations sum to -7.28%, identical to the five-scale estimand. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes.

Alt text: Four smoothed eastern-China maps derived from 121 analysis sites, shown as dots. The spatially centred surfaces show a weaker north–south contrast on high days; the high-minus-middle surface changes from positive in the north to negative in the south, and most observed-node allocations are negative, with the strongest negative values concentrated in the central and southern parts of the domain.

only 18 summers. At broad scales, the nested bases show that latitude accounts for most of the low-dimensional geographic–topographic structure; structured localisation is weaker at 126 km. Node allocations are negative at 106 sites and sum exactly to -7.28%. Each is a symmetric incident-edge allocation.

The climatology–anomaly decomposition in Equation (12) separates the broad-scale change into anomaly energy and climatological alignment. At 126 km, anomaly-energy change contributes -2.63 points and the climatology–anomaly cross term contributes -0.22 points. At 2,013 km, the anomaly-energy component is $+0.41$ points (95% descriptive (t)-reference interval based on between-summer variation, $[-0.50, 1.31]$), whereas the cross term is -13.68 points (95% descriptive (t)-reference interval based on between-summer variation, $[-16.68, -10.67]$) and is negative in all 33 summers. Thus the broad-scale contraction was accounted for algebraically by high-day anomalies opposing the chosen monthly climatological pattern, while their own broad-scale energy changed little.

For each evaluation summer, a sensitivity calculation rebuilt the three monthly climatological fields from the other 34 summers and repeated the exact decomposition. At $h = 2{,}013$ km, this leave-one-summer-out version gave $+0.40$

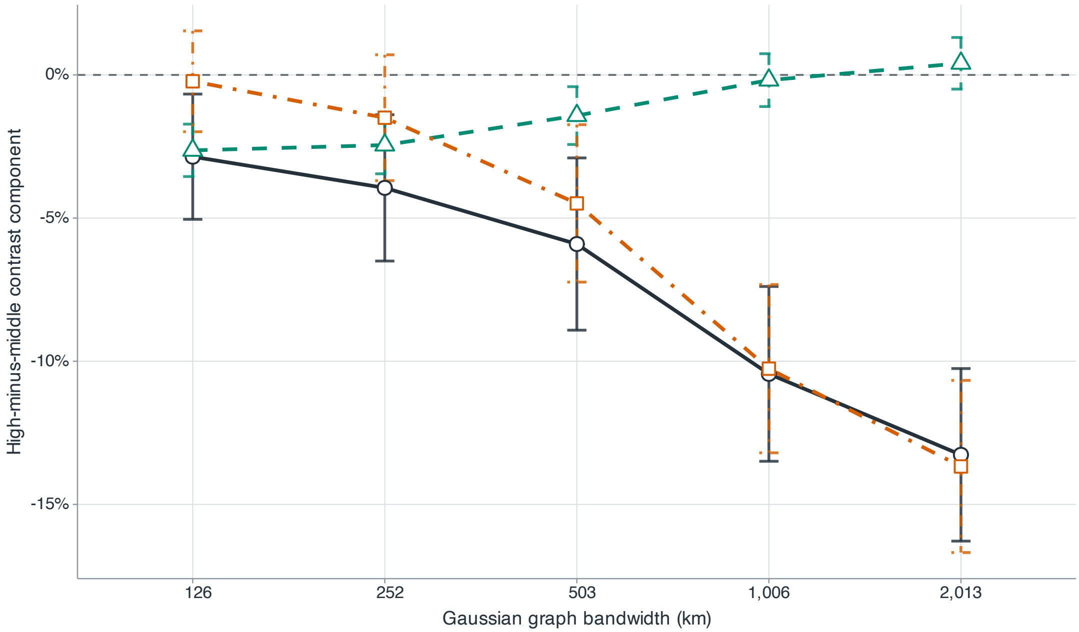

Figure 6 Exact decomposition of the raw-field graph contrast. Dark circles, teal triangles and orange squares denote total change, anomaly-energy change and the climatology–anomaly cross component; bars are 95% descriptive intervals. Monthly-climatology energy cancels within each record.

Alt text: Across five bandwidths, dark circles become increasingly negative. Teal triangles remain near zero at broad scales, while orange squares closely follow the negative total.

percentage points for anomaly energy and -13.67 points for the cross term, compared with $+0.41$ and -13.68 points for the inclusive climatology. The maximum numerical failure of the daily identity was 7.2×10^{-14}. The structural interpretation is therefore insensitive to including the evaluated summer in its climatological reference.

The convergence analysis uses the primary equirectangular distance matrix and compares each grid with the 465-site result using the evaluation events. The 121-site scale profile differs by at most 0.99 percentage points, with a five-scale root mean squared difference of 0.76 points. Latitude refinement reduces these values to 0.50 and 0.38 points; longitude refinement gives 0.93 and 0.73 points. On the 465-site grid, the primary distances and original events give -8.03%; retaining those distances but reselecting events gives -7.25%. Contraction strengthens with bandwidth at every resolution. Agreement between spatial supports is weakest at 126 km, where the 121-site spacing and station support are coarsest.

Using the primary equirectangular distance matrix, cosine-latitude weighting gave -8.11% with the original labels and -7.67% after relabelling. Replacing the distance matrix by WGS84 geodesic distances gave corresponding estimates of -7.80% and -7.37%; the WGS84 equal-site estimate was -6.97%. On the full 465-site domain, WGS84 distances with reselected events gave -6.93%, and relabelled boundary checks ranged from -6.32% to -7.97%. The non-rectangular Natural Earth China-land intersection gave -6.35%, with a scale curve of $-2.09, -2.80, -5.06, -9.64$ and -12.18%. Every support retained the negative direction and strengthening contraction with bandwidth, although the magnitude depends on the specified domain.

The reported -7.28% is an equally aggregated relative change on the squared-difference, or semivariance, scale. It corresponds to a -3.71% RMS change when transformed as $\sqrt{1-0.0728}-1$; averaging record-specific RMS ratios gives -3.97%. At the largest bandwidth, the mean record-level RMS pairwise difference falls from 5.62 to 5.21 °C, a reduction of 0.41 °C; at the smallest bandwidth it falls from 2.31 to 2.27 °C, a reduction of 0.04 °C.

Threshold, Stull and sitewise-maximum sensitivities range from -6.90% to -8.27%. Early- and late-period summaries have overlapping descriptive intervals. With alternative event partitions, upper versus middle tertiles give -6.48%; symmetric 20th/80th and 30th/70th cut points give -8.48% and -6.88%, respectively. All retain stronger contraction at larger bandwidths (Supplement, Section S12). The magnitude depends on the selected states; the negative direction and scale ordering are stable over these choices.

For comparison, the same equal-month, equal-scale and equal-year formula gives -15.14% for the 2015 and 2022 design summers.

5.6. Robustness of the findings

The 495 middle-day denominators range from 2.25 to $25.99\ ^{\circ}\mathrm{C}^2$ across records and scales; within-scale coefficients of variation are 0.076–0.177, with a minimum of $2.25\ ^{\circ}\mathrm{C}^2$. In the heavy-tail simulation, randomisation calibration coexisted with substantially higher raw-ratio RMSE. The back-transformed log-ratio, our principal robustness result, is -8.30%; the bounded symmetric contrast $2(Q^E - Q^M)/(Q^E + Q^M)$ is -8.59%. With the primary equirectangular distances, cosine-latitude weighting gives -8.11% using the original labels and -7.67% after relabelling. With WGS84 distances, the corresponding estimates are -7.80% and -7.37%. Kernels matched to the Gaussian weighted mean distance give -7.71% (exponential) and -7.33% (compact quadratic). All preserve the negative direction and strengthening contraction with bandwidth.

The continuous-intensity diagnostic gives a five-scale $\log Q_h$ slope of $-0.065\ ^{\circ}\mathrm{C}^{-1}$ (95% descriptive (t)-reference interval based on between-summer variation, $[-0.082, -0.048]$), with negative summer slopes in 31 of 33 years. Scale-specific slopes become more negative from $-0.032\ ^{\circ}\mathrm{C}^{-1}$ at 126 km to $-0.110\ ^{\circ}\mathrm{C}^{-1}$ at 2,013 km; the corresponding negative-year counts rise from 29 to 32. The negative association extends across the within-month WBT distribution.

The estimated contrast depends on the definition of the spatial field. Subtracting each site's 35-year monthly climatology reduces the estimate to -3.92% ($[-8.67, 0.83]\%$); subtracting the site-year-month mean gives -1.32% ($[-5.63, 2.99]\%$). Site-month standardisation yields -9.87%, but targets a different dimensionless field. Together with the gradient decomposition, these results show that high-day anomalies offset the cross-site climatological gradient, while their own broad-scale energy changes little.

The regime dates have a systematic but nonmonotone seasonal pattern. Averaged over June–August, high days occur 1.96 days later than middle days (95% descriptive (t)-reference interval based on between-summer variation, $[1.10, 2.83]$ days): they occur 7.82 and 6.34 days later in June and July, but 8.29 days earlier in August. Removing a separate linear day-of-month trend within each site–year–month gives -8.81% ($[-13.43, -4.19]\%$); subtracting the leave-one-year calendar-day climatology gives -9.81% ($[-14.22, -5.40]\%$). The negative contrast persists after both seasonal adjustments, although the weaker site-month and site–year–month anomaly results show that different demeaning operations target different spatial fields.

Calendar matching attenuates the contrast but preserves its direction. With 95% descriptive (t)-reference intervals based on between-summer variation in parentheses, the ± 3-day estimator is -5.21% ($[-8.01, -2.41]\%$), with 24 of 33 summer contrasts negative and 670 of 792 high days matched; the ± 5-day estimator is -5.72% ($[-8.43, -3.02]\%$), with 25 negative summers and 750 matched high days. Every one of the 99 month-year records has at least one eligible match under both windows. The profiles run from -2.32% to -8.87% for ± 3 days and from -2.54% to -9.89% for ± 5 days, retaining the strengthening contraction with bandwidth.

All fixed-hour and daily-mean analyses yielded negative estimates. Using the original event days, all 24 fixed-hour estimates are negative, from -10.07% at 00 UTC to -6.03% at 11 UTC, and daily-mean fields give -7.58%. When labels are recomputed from each fixed hour, all estimates remain negative, from -14.65% at 22 UTC to -5.42% at 11 UTC; labels recomputed from daily-mean WBT give -9.14%. Peak hours have nearly identical distributions in the original high and middle groups (median 06 UTC). Figure 7 compares the main estimate with transformation, graph-construction, spatial-resolution and field-definition alternatives.

The station comparison contains 175,172 exact-hour matches at 28 usable stations. ERA5-Land minus station WBT has a bias of -0.263 °C, a mean absolute error of 0.938 °C and a root mean squared error of 1.270 °C. The within-station centred correlation is 0.916. The two offshore stations excluded from these summaries had no finite values at their nearest ERA5-Land grid cells.

At Gaussian bandwidths 126, 252, 503, 1,006 and 2,013 km, the station high–middle contrasts are $-6.23, -7.23, -12.50, -18.03$ and -21.47%; the corresponding ERA5-Land values on identical station subsets are $-11.60, -12.74, -17.13, -21.60$ and -24.23%. The numbers of negative station summers across the five scales are 6, 7, 8, 9 and 9. Averaged over the three predesignated station-supported scales, the station contrast is

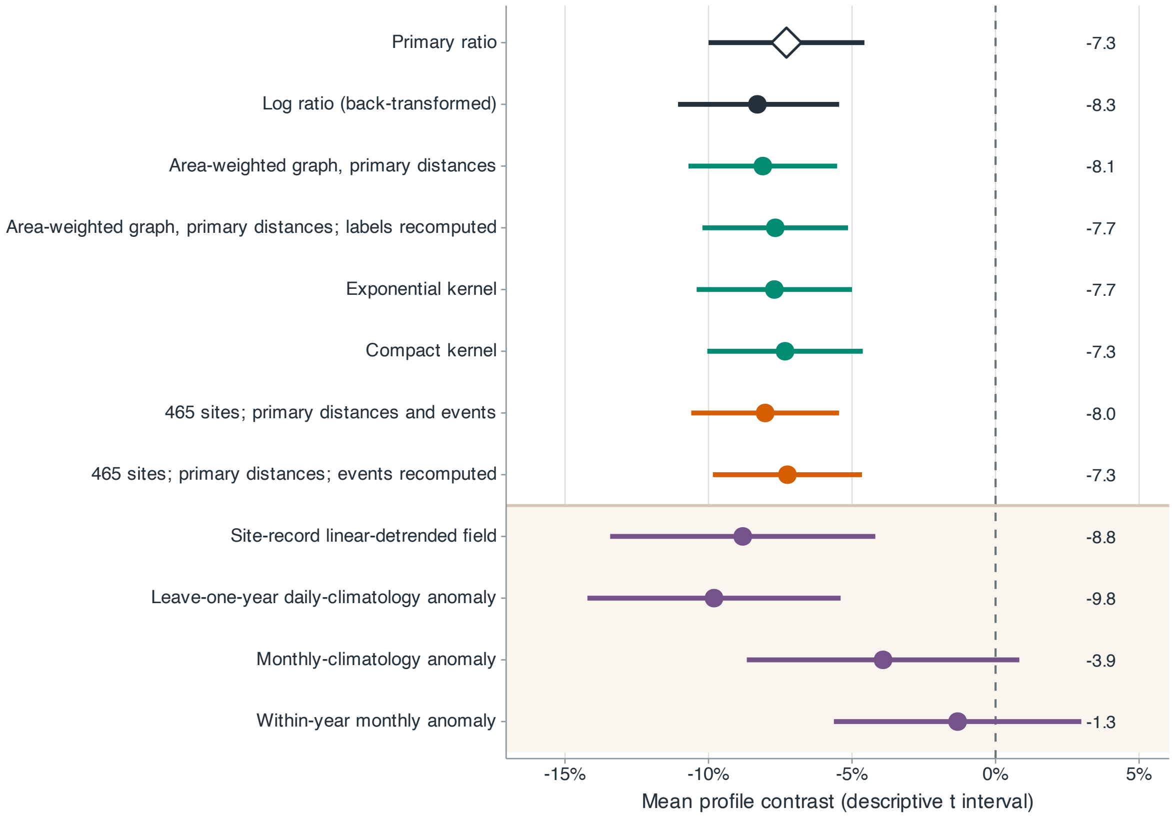

Figure 7 Robustness of the mean profile contrast. Dark rows denote the main and transformation analyses; green, orange and purple rows denote graph construction, spatial resolution and field definition. Points and bars are estimates and descriptive t-reference intervals based on between-summer variation.

Alt text: A forest plot shows mostly overlapping negative estimates across transformation, graph and resolution choices. Two purple-shaded anomaly-field rows lie closest to zero.

-17.34% ($[-28.49, -6.18]\%$), with 9 of 10 summer contrasts negative; the paired ERA5-Land contrast is -20.99% ($[-28.57, -13.40]\%$), with all 10 negative. The station and matched reanalysis estimates agree in direction at the three designated broad scales. At 126 km, both intervals include zero and provide no clear evidence of a local-scale contrast (Figure 8).

Of 240 scheduled high and 440 middle fields, 122 and 223, respectively, had at least 10 stations. The high-minus-middle differences in mean station count and eligible-field fraction were 0.16 sites (descriptive interval $[-2.31, 2.62]$) and 0.002 ($[-0.095, 0.098]$). The three-band station contrast was -18.08% on a fixed common support within each summer. Requiring at least two, three or five retained days per regime gave -13.42, -15.24 and -14.56%; the five-day interval included zero because only six summers remained. When regional means over the available station network defined both the daily peak and event labels, the contrast was -15.99% ($[-22.95, -9.02]\%$). Thus the broad direction survives the fixed-support check and a station-based event definition on the available dynamic network, but record sparsity limits precision.

Station elevations ranged from 3 to 2,063 m. In the WGS84 station calculation, treating sea-level pressure as surface pressure instead of applying the elevation conversion changed WBT by 0.072°C on average in absolute value (maximum 1.346°C) and the three-band station contrast from -17.37% to -17.29%.

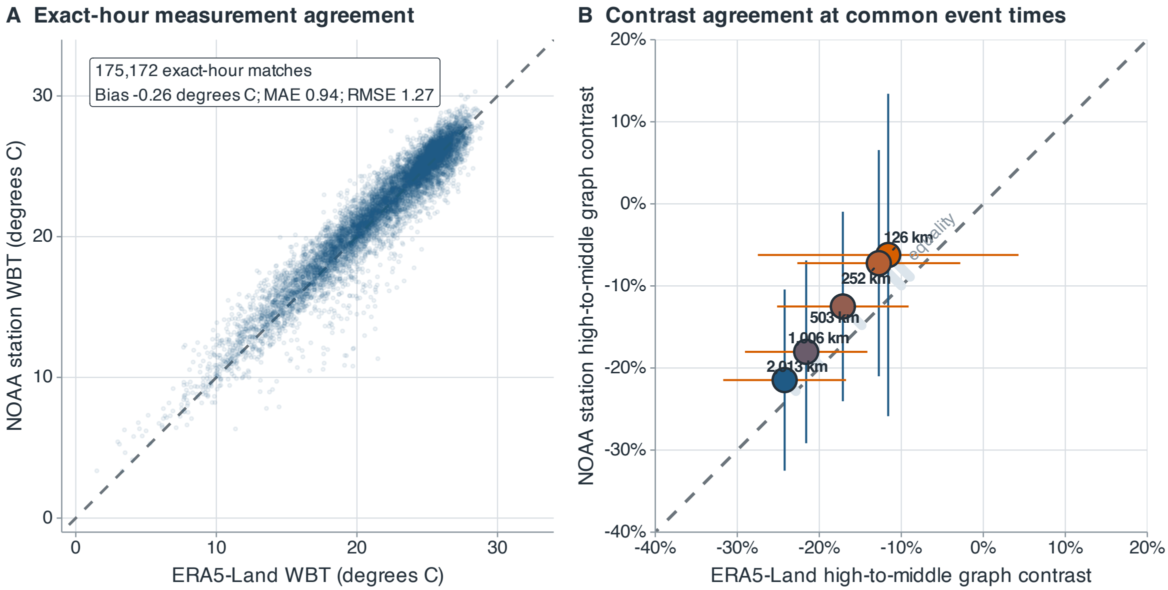

Figure 8 NOAA station and ERA5-Land agreement in ten station-comparison summers. (A) Exact-hour WBT at 28 stations, with summaries from all 175,172 matches. (B) High–middle graph contrasts at the five bandwidths, using ERA5-Land peak times and labels and identical station subsets. Dashed diagonals denote equality. Horizontal and vertical bars in panel B are paired descriptive t-reference intervals based on between-summer variation.

Alt text: Two panels compare NOAA stations with ERA5-Land. Exact-hour WBT points in the left panel cluster around the equality line. In the right panel, station and ERA5-Land high–middle contrasts are negative at all five bandwidths and become more negative as the bandwidth increases; station estimates are slightly closer to zero.

6. Discussion

High regional wet-bulb temperature in eastern China accompanies a flatter north–south field. Across the 33 evaluation summers, the five-scale mean squared pairwise contrast was 7.28% lower on high-mean days, with negative contrasts in 28 summers. Contraction increased from 2.86% at a 126-km bandwidth to 13.27% at 2,013 km. At the larger bandwidth, mean record-level RMS pairwise differences fell from 5.62 to 5.21 $^\circ$C. The 1950–1990 extension showed the same scale ordering, with a mean contrast of -11.04% and negative values in 40 of 41 summers. NOAA observations also showed contraction at 503 km and broader, averaging -17.34% over these three bandwidths.

The decompositions identify the geographic structure of this change. After subtracting each day's regional mean, high-day fields are warmer in the north and cooler in the south than middle-day fields. The north–south component accounts for most of the contraction. At broad scales, daily anomalies oppose the monthly climatological gradient, while their own spatial energy changes little. The attenuation after site-month and site–year–month demeaning further associates the result with the cross-site climatic gradient. Thus the regional variance decreases chiefly through weaker broad geographic contrasts, alongside smaller changes over short distances.

The coverage analysis gives these field changes a geographic magnitude. At 24°C, high-day exceedance area is about 416,000 km^2 greater than on middle days, and the largest connected exceedance region is about 405,000 km^2 larger. Average differences peak near 23–24°C and remain positive across the examined range. Regional mean and spatial pattern change together in this comparison. Their combined information is well captured by a regression using the mean, variance, signed latitude gradient and direct 121-site exceedance proportion. Adding graph scales reduces integrated coverage errors by less than 0.01 percentage points for area and about 0.01 points for connected area. Binned semivariances give comparable or slightly smaller errors. The graph profile contributes a smooth scale comparison and an exact spatial decomposition, while the strong simple baseline retains most information about the coverage states.

Spatial resolution and record length affect the precision of these results. The median nearest-neighbour distance is 150 km, and station intervals at the 126-km bandwidth include zero. Boundary changes preserve the scale ordering while changing the average contrast. The historical extension shares the ERA5-Land reanalysis system, including preliminary forcing in 1950–1978. Coverage areas are approximated on the $0.8^\circ \times 0.9^\circ$ lattice; coastal cells and narrow connections depend on this resolution. The 465-site reference shares the reanalysis system and contains the original 121 sites. The additional-site comparison evaluates area at the other 344 sites. NOAA coverage spans 28 stations and ten summers; requiring more days per regime reduces the available sample. These features favour interpretation at broad spatial scales over the analysed domain. For inference, cyclic shifts use the conditional invariance assumption in Equation (14); the simulations quantify sensitivity to cross-record dependence and the undercoverage of Student and HAC intervals in short, correlated summer sequences.

The daily contrast concerns the spatial organisation of humid heat during regional warm states. Long-term changes in China's north–south wet-bulb temperature pattern have also been associated with heterogeneous water-vapour changes (Wang et al., 2024). Relating the daily pattern to circulation and moisture transport would connect these timescales. Here, the observed broad-scale contraction accompanies larger simultaneous humid-heat regions, and the direct coverage measurements establish the size and temperature range of that association.

Data availability

ERA5-Land is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land`). The NOAA ISD station archive (Smith et al., 2011) is available at `https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database`. Code and retained results for this version are archived at `https://github.com/Stork343/eastern-china-wet-bulb-contraction/releases/tag/jrssc-submission-v6-2026-09-08`. The accompanying `jrssc_reproducibility_bundle.zip` contains the current manuscript, analysis code, protocols, site and station manifests, and retained results, including the September coverage comparison. Its `code/revision_202609/README.md` gives the application-analysis entry points; `output_revision_application/` contains the fixed specification, coverage curves, held-out predictions and annual comparisons. Provider-data download and field-construction programs are included. The raw reanalysis and station archives are obtained from the providers above.

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Conflict of interest

The authors declare no conflict of interest.

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Table 1. Held-out mean absolute error in area percentage points. Errors are averaged over all 77 thresholds (9.5–28.5°C at 0.25°C steps), then equally over months and 33 evaluation summers. All-day errors use 3,036 June–August daily fields in 1991–2025 excluding 2015 and 2022; high-day errors use the 792 original upper-quartile days. The last two columns use the 344 additional sites. All inputs come from the original 121 sites.

	Exceedance area		Largest component		Additional sites	
Method	All	High	All	High	All	High
Mean and season	2.794	2.291	3.070	2.593	2.818	2.314
Strong simple baseline	0.827	0.747	1.389	1.268	1.028	0.929
Simple + variogram	0.815	0.741	1.366	1.266	1.013	0.922
Simple + graph	0.820	0.743	1.379	1.265	1.020	0.925
Spatial interpolation	1.398	1.414	1.823	1.859	1.861	1.882

Table 2. Physical scale of the evaluation-period graph contrasts. h is the Gaussian bandwidth. $\bar{d}_w$ is the edge-weighted mean pair distance, and N_{eff} is the Kish effective edge count. Graph contrasts are relative changes in weighted squared differences. RMS contrasts average record-specific ratios of $(2Q)^{1/2}$; Negative is the number of negative summer graph contrasts.

h (km)	$\bar{d}_w$ (km)	N_{eff} (edges)	Graph contrast (%)	RMS contrast (%)	Negative (/33)
126	200	366	-2.86	-1.54	22
252	319	1,138	-3.94	-2.14	23
503	549	3,223	-5.91	-3.25	25
1,006	826	6,005	-10.44	-5.70	29
2,013	992	7,109	-13.27	-7.22	30